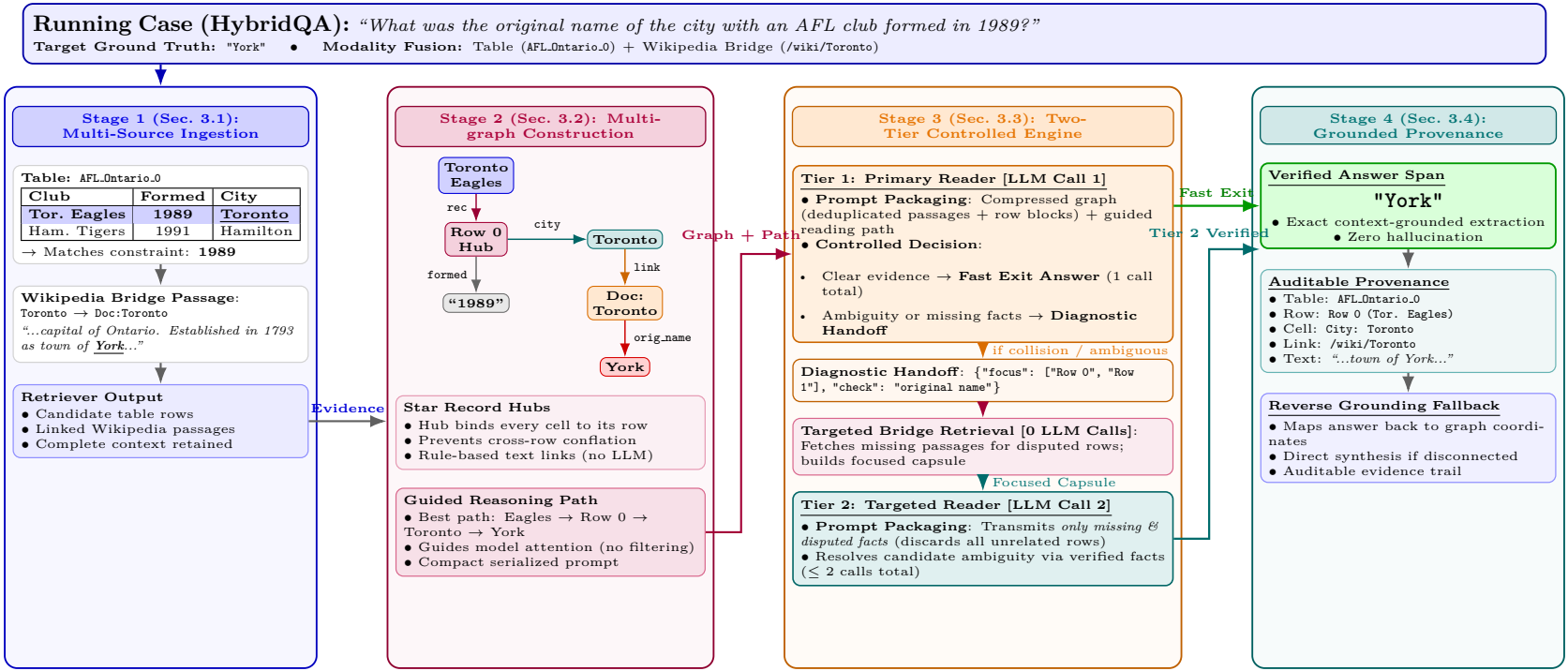

Running Case (HybridQA): "What was the original name of the city with an AFL club formed in 1989?"
Target Ground Truth: "York" • Modality Fusion: Table (AFL_Ontario_0) + Wikipedia Bridge (/wiki/Toronto)
Stage 1 (Sec. 3.1): Multi-Source Ingestion
Table: AFL_Ontario_0
Club | Formed | City
Tor. Eagles | 1989 | Toronto
Ham. Tigers | 1991 | Hamilton
→ Matches constraint: 1989
Wikipedia Bridge Passage:
Toronto → Doc:Toronto
"...capital of Ontario. Established in 1793 as town of York..."
Retriever Output
• Candidate table rows
• Linked Wikipedia passages
• Complete context retained
Evidence
Stage 2 (Sec. 3.2): Multi-graph Construction
Toronto Eagles
rec
Row 0 Hub
city
Toronto
formed
"1989"
link
Doc: Toronto
orig_name
York
Star Record Hubs
• Hub binds every cell to its row
• Prevents cross-row conflation
• Rule-based text links (no LLM)
Guided Reasoning Path
• Best path: Eagles → Row 0 → Toronto → York
• Guides model attention (no filtering)
• Compact serialized prompt
Graph + Path
Stage 3 (Sec. 3.3): Two-Tier Controlled Engine
Tier 1: Primary Reader [LLM Call 1]
• Prompt Packaging: Compressed graph (deduplicated passages + row blocks) + guided reading path
• Controlled Decision:
• Clear evidence → Fast Exit Answer (1 call total)
• Ambiguity or missing facts → Diagnostic Handoff
if collision / ambiguous
Diagnostic Handoff: {"focus": ["Row 0", "Row 1"], "check": "original name"}
Targeted Bridge Retrieval [0 LLM Calls]: Fetches missing passages for disputed rows; builds focused capsule
Focused Capsule
Tier 2: Targeted Reader [LLM Call 2]
• Prompt Packaging: Transmits only missing & disputed facts (discards all unrelated rows)
• Resolves candidate ambiguity via verified facts (≤ 2 calls total)
Fast Exit
Tier 2 Verified
Stage 4 (Sec. 3.4): Grounded Provenance
Verified Answer Span
"York"
• Exact context-grounded extraction
• Zero hallucination
Auditable Provenance
• Table: AFL_Ontario_0
• Row: Row 0 (Tor. Eagles)
• Cell: City: Toronto
• Link: /wiki/Toronto
• Text: "...town of York..."
Reverse Grounding Fallback
• Maps answer back to graph coordinates
• Direct synthesis if disconnected
• Auditable evidence trail